

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 03
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Title: Frequency Modulation

Objectives:

1. To generate Frequency Modulated wave using IC 8038 function generator
2. To find frequency deviation

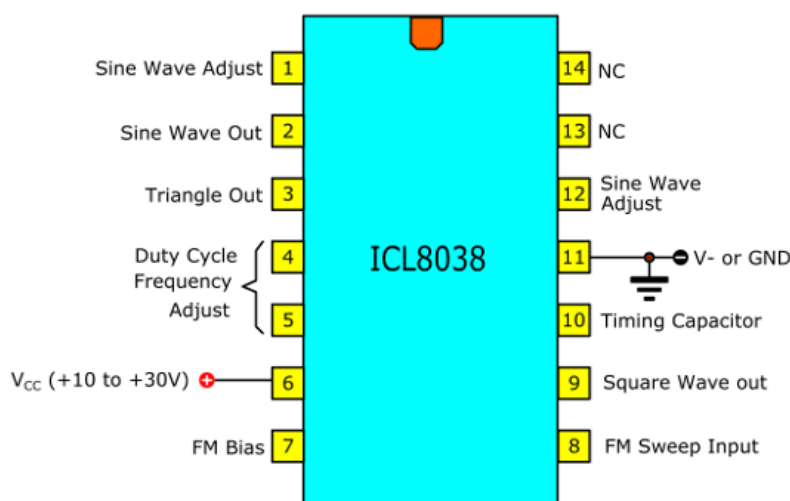
OUTCOME:

1. analyze and compare analog modulation schemes.
2. understand noise effect on the performance of communication systems.

• **Procedure:**

- Connections are made as shown in the circuit diagram.
- By switching off the modulating signal, note down the carrier frequency (f_c) at pin2 of IC 8038.
- Apply modulating signal of suitable amplitude to get undistorted FM.
- Measure the modulation index of the generated FM signal using CRO and observe the corresponding spectrum and note the number of sidebands and their respective powers.
- Vary the modulating frequency and modulating voltage. Note the number of sidebands and their powers on spectrum analyser.
- Plot the time domain FM signal (i.e. signal observed on CRO) on the graph paper.

• **Pin diagram of IC 8038** (Image Source: <https://www.eleccircuit.com/mini-function-generator-circuit-using-icl8038/>)



• **Circuit diagram: Draw circuit diagram**

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• **Observation Table:**

1) Free running carrier frequency, $f_c = \underline{\hspace{2cm}}$

2) For constant modulating frequency, $f_m = \underline{\hspace{2cm}}$

$V_{m(p-p)}$	T_{min}	T_{max}	Δf	$m_f = \Delta f / f_m$

3) For constant amplitude of modulating signal $V_{m(p-p)} = \underline{\hspace{2cm}}$

$V_{m(p-p)}$	T_{min}	T_{max}	Δf	$m_f = \Delta f / f_m$

- **Calculations: Show calculations for Δf and m_f**
- **Waveforms: Sketch neat and labelled modulating signal, carrier signal and frequency modulated wave with proper scale.**
- **Post lab activity:**
 - Prepare a one page (strictly) table/chart having following details:
 - FM frequency bands (start/end frequency, bit rate, frequency range etc.)
 - FM radio stations in Mumbai (number of stations, name, frequency bands, history, bitrate, other technical details)

Signature of faculty in-charge